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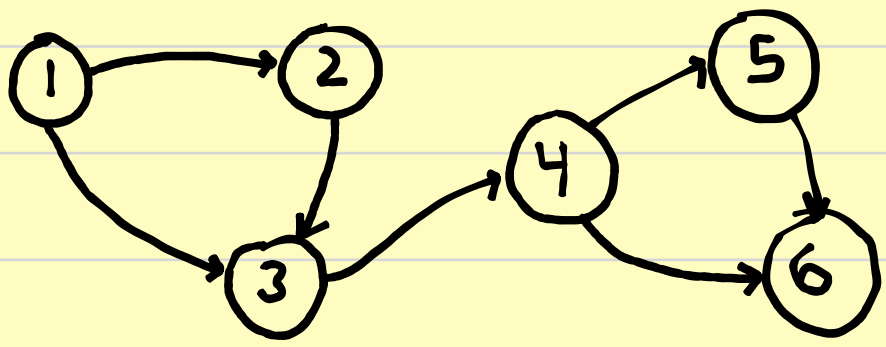
JAVIER

H 4

NETWORKS

A Network is defined by a group of nodes and edges called GRAPH.

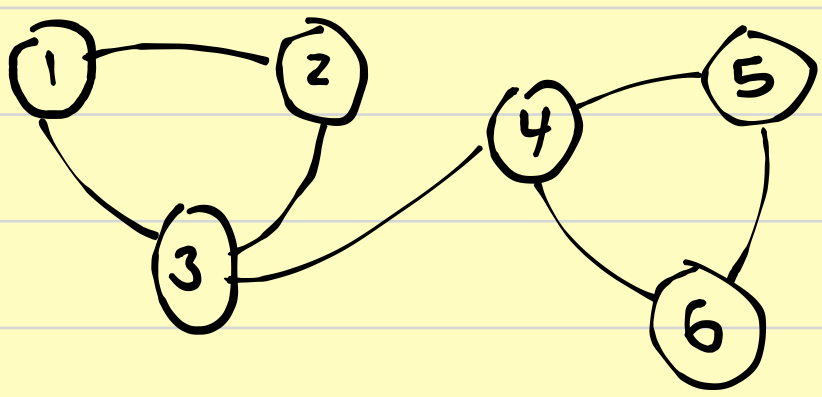
GRAPH ≡ G(Nodes ∪ Edges) ≡ G(N, E)



Directed Network $\xrightarrow{E}$

E((1→2), (2→3), (3→4), (1→3), (4→5), (5→6), (4→6))

N(1,2,3,4,5,6)



Undirected Network $\overset{E}{\text{———}}$

E((1—2), (2—3), (3—4), (4—5), (1—3), (5—6), (4—6))

N(1,2,3,4,5,6)

How do we quantify networks?

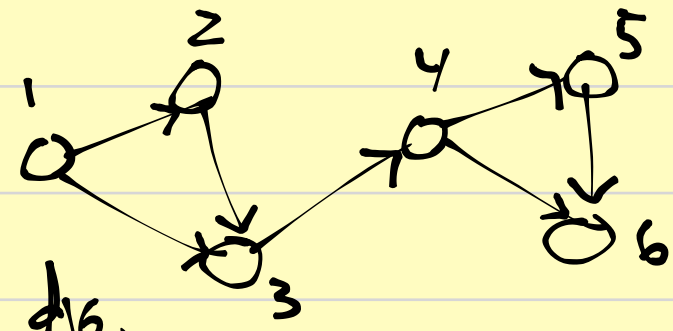
- AVERAGE PATH LENGTH (APL)
- CLUSTERING COEFFICIENT (CC)
- DEGREE DISTRIBUTION (DD)

. LAPLACE MATRIX (LM)

. FIEDLER VEKTOR (FV)

1. APL . Average distance between nodes.

$$\boxed{APL = \frac{1}{N(N-1)} \cdot \sum_{i,j=1}^{n} d_{ij}}$$



$$APL = \frac{1}{6(6-1)} \cdot \Big[\overset{d_{12}\ d_{13}\ d_{14}\ d_{15}\ d_{16}}{① (1+1+2+3+3)} + \overset{d_{21}\ d_{23}\ d_{24}\ d_{25}\ d_{26}}{② (0+1+2+3+3)} + ③ (0+0+1+2+2) + ④ (0+0+0+1+1) + ⑤ (0+0+0+0+1) + ⑥ (0+0+0+0+0) \Big] = \cdots =$$

The best APL is the shortest.

If you have 2 networks with the same number of nodes, you may compare their APL. The one that has the smallest, will probably perform better.

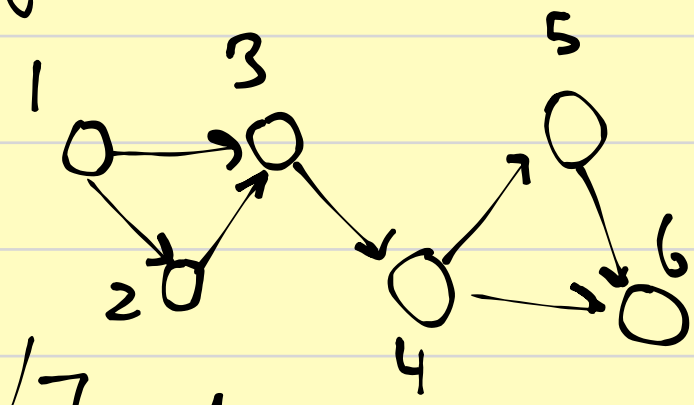
2. CC . Describes how good groups are created in the network.

$$\boxed{CC = \frac{1}{N} \sum_{i=1}^{N} \frac{2L_i}{K_i(K_i-1)}}$$

$L_i \equiv$ number of relationships between the neighbours of node "i".

$K_i \equiv$ number of neighbours of node "i". $\equiv$ DEGREE

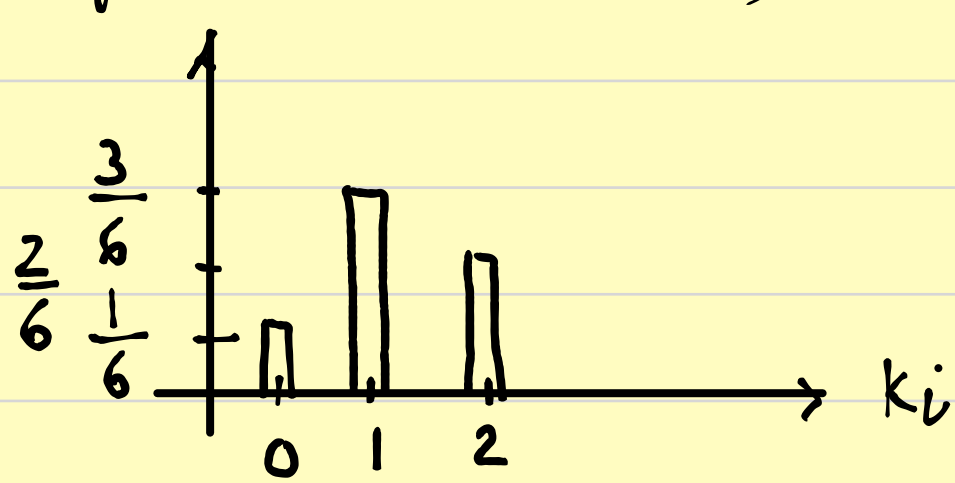
$$CC = \frac{1}{6}\left[\overset{①}{\frac{2 \cdot \overset{L_1}{1}}{2 \cdot (2-1)}} + \overset{②}{\frac{2 \cdot \cancel{0}}{}} + \overset{③}{} + \overset{④}{\frac{2 \cdot \overset{L_4}{1}}{2(2-1)}} + \overset{⑤}{} + \overset{⑥}{}\right] = \frac{1}{3}$$

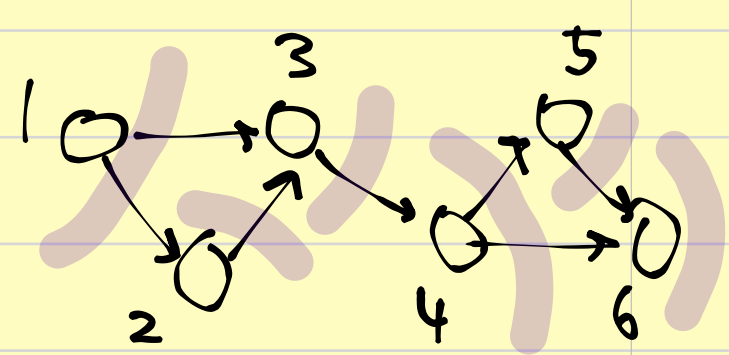


The higher the CC, the better connected are the groups in the network.

You can compare two networks with the same number of nodes by comparing the CC.

3. Degree distribution (DD)



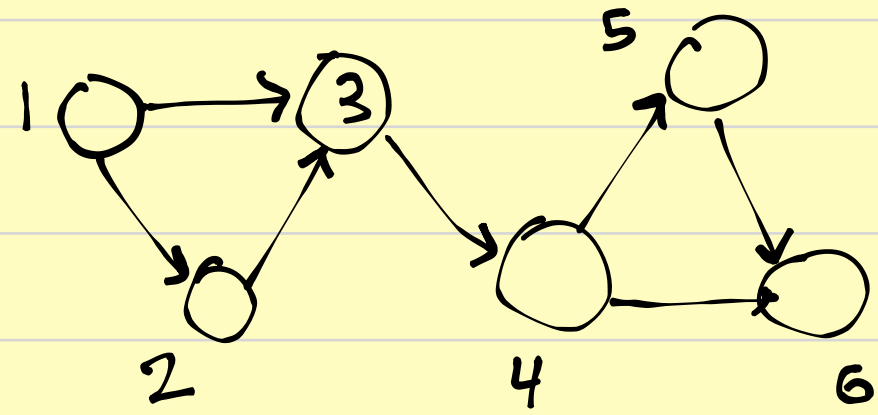


4. Laplacian Matrix (LM)

It contains all relevant info of the network.

$$\boxed{L = D - A}$$

$$D = \begin{cases} k_i & i = j \\ 0 & i \neq j \end{cases} \qquad A = \begin{cases} 1 & \text{if connection} \\ 0 & \text{otherwise} \end{cases}$$



$$L = \begin{bmatrix} 2 & & & & & \emptyset \\ & 1 & & & & \\ & & 1 & & & \\ & & & 2 & & \\ & & & & 1 & \\ \emptyset & & & & & 0 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} =$$

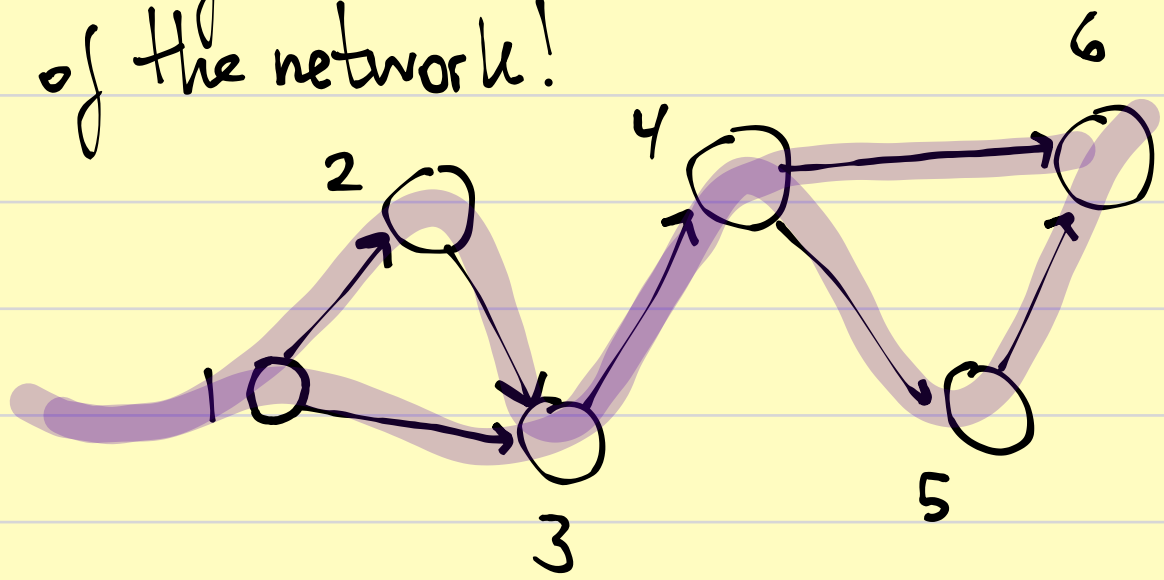
$$= \begin{bmatrix} 2 & -1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 2 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

5. Fiedler vector.

The second eigenvector of the laplacian matrix.
The fiedler vector has a sign structure.

$$
Fv = \begin{array}{c} 1 \\ 2 \\ 3 \\ \vdots \\ \vdots \\ \\ n \end{array} \begin{bmatrix} -1 \\ -1 \\ -1 \\ 0 \\ 0 \\ +1 \\ +1 \\ +1 \end{bmatrix}
$$

the first node at which the sign changes indicates the bottleneck of the network!



Barábasi . „Complex Networks"